

Harsharandeep Dhillon

Hayward, CA | [LinkedIn](#) | 510-634-2927 | sharandhillon8123@gmail.com

EDUCATION

Chabot College

Associate of Science in Computer Science (GPA: 3.9)

Hayward, CA

May 2025

- Organizations/Awards: NSF Stem scholarship, Academic Honors List, Women In Tech Club - Vice President
- **Incoming Junior Transfer student - Applied Mathematics at UC Berkeley, Fall 2025**

WORK EXPERIENCE

AI/ML fellow at Cornell Tech: Selected through Break Through Tech; currently completing a machine learning course led by Cornell Tech to build applied skills in model development, evaluation, and real-world AI use cases. *May 2024 – present*

UC Berkeley - College of Engineering

Program Assistant - Girls in Engineering Program

Berkeley, CA

June 2025 – Present

- Support instructors in facilitating sessions on 3D printing, coding, and circuitry while ensuring all students stay involved.
- Assist with hands-on STEM workshops for middle school girls, introducing engineering topics through design challenges.
- Work closely with staff to improve workshop delivery and provide mentorship that encourages curiosity and confidence in STEM.
- Adapt explanations to meet different experience levels, helping create a welcoming, inclusive environment for all participants.

UL Solutions

Test Engineering Intern

Fremont, CA

June 2024 – August 2024

- Gained hands-on experience in designing, building, and testing RF systems, including power amplifiers, up/down converters, filters, oscillators, and transmission lines.
- Developed Python scripts to generate and analyze signals for circuit testing, reducing debugging time by 20% and improving overall testing efficiency.
- Collaborated with cross-functional engineering teams to define and achieve technical specifications, enhancing project execution and problem-solving efficiency.
- Enhanced safety protocols in lab environments, leading to a 30% reduction in workplace incidents.
- Applied RF concepts like impedance matching, filter design, and circuit optimization, ensuring good compliance with engineering standards.

Chabot College

Teaching Assistant and Math Tutor

Hayward, CA

January 2025 – May 2025

- Supported students in understanding core math concepts such as algebra, calculus, and differential equations through step-by-step problem-solving and real-world examples.
- Clarified complex topics by providing structured guidance on assignments, helping students build confidence in foundational and advanced math skills.
- Encouraged critical thinking by breaking down challenging problems into manageable steps and fostering analytical reasoning.
- Created an inclusive and collaborative learning environment by adapting explanations to different learning styles.

PROJECTS

Airbnb Price Prediction Model

Machine Learning Coursework Project

- Developed a regression-based machine learning model to predict Airbnb listing prices using real-world San Francisco data, with a focus on data cleaning, outlier handling, and one-hot encoding for categorical variables..
- Applied and compared multiple models including Linear Regression, Decision Trees, and Ridge Regression, while tuning hyperparameters to optimize RMSE and increase predictive accuracy through cross-validation and model evaluation.

Battery Degradation Analysis from Fast Charging

Stanford Small Science Group Research Project

- Investigated the impact of frequent fast charging on lithium-ion battery degradation using capacity-cycle data to model long-term performance trends.
- Visualized state-of-health (SoH) changes and built data-driven charts to highlight early signs of capacity fade. Worked under mentorship from a Stanford researcher to gain insights into electric vehicle battery research and energy system sustainability.

Extracurricular Activities and Skills

Skills: Microsoft Office (Word, Excel, PowerPoint) | Python | JavaScript | HTML5 | CSS3 | React.js | scikit-learn | C++ | Pandas | SQL | GitHub | Linux | Windows | macOS | Strong problem-solving | effective communication | collaboration | time management | attention to detail | adaptability | critical thinking.

Course work: Physics - Electricity and Magnetism | Newtonian Mechanics | Data Structures and Algorithms | Object-Oriented Programming in C++ | Calculus | Elementary Linear Algebra | Elementary Differential Equations | Introduction to Linux and Unix Systems | Assembly Language | Discrete Mathematics |